

Day 49 - Port Security

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Port Security Info

Port Security General Information

Port Security is a feature of Cisco switches that allows you to control which **source** MAC address(es) are allowed to enter the switchport.

- If an **unauthorized source** MAC address enters the port, an action will be taken.
- The default action is to place the interface in an **'err-disabled'** state.
- When you enable **port security** on an interface, **one** MAC address is allowed by default.
 - You can configure the allowed MAC address manually.
 - The allowed MAC addresses can be configured manually or dynamically. A combination of both is also possible.
 - If not configured manually, the switch will allow the **first source** MAC address that enters the interface.
 - You can also change the maximum number of MAC addresses allowed.

Reasons to Use Port Security

Port Security allows network admins to control which devices are allowed to access the network.

- Despite MAC address **spoofing** being a simple task and commonly used to deploy attacks on networks, **port security**'s main benefit is the ability to limit the number of MAC addresses allowed on the interface. This is beneficial as it can limit the amount of attacks that can be deployed on the network.

General Configuration - Port Security

- **show port-security**
 - Shows general information about the current **port security** configuration
 - Includes information like:
 - Ports that have **port security** enabled
 - Max Secure MAC addresses
 - Currently configured MAC addresses
 - Current security violations
 - Configured security action
 - Total addresses in the system
 - Max addresses limit in the system
- **show port-security interface INTERFACE-ID**
 - Shows **port security** information of the specified interface.
 - Includes information such as:
 - **Port security** status
 - Port status
 - Aging time (for static MAC addresses)
 - Aging type (for static MAC addresses)
 - Status of Secure Static MAC address aging
 - Maximum MAC addresses allowed
 - Total MAC addresses (**Dynamic** and **Static** addresses)
 - Configured MAC addresses (Manually configured)
 - Sticky MAC addresses (**Sticky Dynamic** addresses)
 - Last source MAC address + VLAN
 - Security violation count
- **show errdisable recovery**
 - Shows information about the current **'err-disable recovery'** configuration
 - Includes information such as:
 - Every disabled/enabled protocol

- Timer interval
- `show mac address-table secure` ^show-mac-addressstable-secure
 - Shows all configured Secure MAC addresses on the switch.
 - Includes information such as:
 - VLAN
 - The Secure MAC address
 - Secure MAC address type
 - Port
- `switchport port-security`
 - Enables **port security** on the current interface.
 - The interface must be in **access** or **trunk** mode for the command to work. If it's **dynamic** mode, the switch will reject the command.
- `switchport port-security violation VIOLATION-MODE { SHUTDOWN | RESTRICT | PROTECT }`
 - Configures the **violation mode** for the **port security** enabled interface
- `switchport port-security mac-address sticky` ^sw-ps-mac-sticky
 - Enables 'Sticky' Secure MAC address learning for **port-security**
- `switchport port-security mac-address sticky MAC-ADDRESS`
 - Manually adds a 'Sticky' Secure MAC address for **port-security**
- `switchport port-security mac-address MAC-ADDRESS`
 - Manually configures a Secure MAC address for **port security**.
- `switchport port-security maximum MAXIMUM-MAC-ADDRESSES`
 - Changes the maximum amount of allowed Secure MAC addresses on the **port security**-enabled interface
- `switchport port-security aging static` ^sw-ps-age-static
 - Enables aging for **port security** Secure Static MAC addresses
- `switchport port-security aging time MINUTES` ^sw-ps-age-time
 - Configures the **port security** Secure MAC address aging time
- `switchport port-security aging type TYPE { ABSOLUTE | INACTIVITY }` ^sw-ps-age-type
 - Configures the **port security** Secure MAC address aging type

Re-Enabling an Interface (Manually)

An **err-disabled** interface can be manually re-enabled by running the `shutdown` command followed by the `no shutdown` command.

Re-Enabling an Interface (ErrDisable Recovery)

- `errdisable recovery cause psecure-violation`
 - Enables err-disable recovery for **port security** violations
- `errdisable recovery interval SECONDS`
 - Determines how long the switch should wait before enabling an err-disabled interface (if the err-disable recovery for that cause is enabled)

Watch out: ErrDisable Recovery and Manual Recovery

Make sure to disconnect any **unauthorized** devices before enabling ErrDisable recovery or manually re-enabling disabled interfaces.

- If you manually configured a MAC address for **port security** and the **unauthorized** device is still connected, it will lead to the interface getting **err-disabled** again.
- If the **port security** system is dynamically learning MAC addresses, then it will clear its dynamic MAC addresses when the interface is disabled; and when the interface becomes enabled again, the **unauthorized** device might become the new Secure MAC address on the interface.

Port Security - Violation Modes

There are three different port security **violation** modes that determine what the switch will do if an **unauthorized** frame enters an interface configured with port security.

1. Shutdown

- Discards traffic from **unauthorized** MAC addresses.
 - Shuts down the port by putting it in an **err-disabled** state.
 - Generates a **Syslog** and/or **SNMP** message when the interface is disabled.
- (Violation counter is set to 1 when the interface is disabled. Resets when the interface is enabled.)*

- violation counter is set to **1** when the interface is disabled. (resets when the interface is enabled)

2. Restrict

- Discards traffic from **unauthorized** MAC addresses.
- The interface is NOT disabled.
- Generates a **Syslog** and/or **SNMP** message each time an **unauthorized** MAC is detected.
- Violation counter is incremented by **1** for each **unauthorized** frame.

3. Protect

- Discards traffic from **unauthorized** MAC addresses.
- The interface is NOT disabled.
- It does NOT generate **Syslog/SNMP** messages for **unauthorized** traffic.
- It does NOT increment the violation counter.

Secure MAC Address Aging

By default, Secure MAC addresses will not age out/expire (**Aging Time: 0** mins).

You can enable Secure **Static** MAC address aging with [switchport port-security aging static](#)

- There are two types of aging for **port security**:

1. Absolute

- After the Secure MAC address is learned, the aging timer starts and the MAC is removed after the timer expires, even if the switch continues receiving frames from that **source** MAC address.

2. Inactivity

- After the Secure MAC address is learned, the aging timer starts counting down but is reset every time a frame from that **source** MAC address is received on the interface.
- They can be configured with the [switchport port-security aging type](#) command.
- The timer can also be configured with [switchport port-security aging time](#).

Sticky Secure MAC Addresses

Sticky Secure MAC addresses are dynamically-learned Secure MAC addresses that will be added to the running-config and will never age out.

- It can be enabled with [switchport port-security mac-address sticky](#)
- Once enabled, dynamically-learned MAC addresses will be added to the running-config like this:
 - [switchport port-security mac-address sticky MAC-ADDRESS](#)
 - (You will need to save the running-config to make them permanent)
- When you issue the [switchport port-security mac-address sticky](#) command, all current dynamically-learned Secure MAC addresses will be converted into **Sticky** Secure MAC addresses.
- If you issue the [no switchport port-security mac-address sticky](#) command, all current **Sticky** Secure MAC addresses will be converted to regular dynamically-learned Secure MAC addresses.

MAC Address Table

Secure MAC addresses will be added to the MAC address table like any other MAC address.

- You can view all secure MAC addresses with the [show mac address-table secure](#) command.
- **Sticky** and **Static** Secure MAC addresses will have a type of **STATIC**.
- Dynamically-learned Secure MAC addresses will have a type of **DYNAMIC**.